8086/8088 Pins

MICROPROCESSORS & INTERFACING (CS F241)

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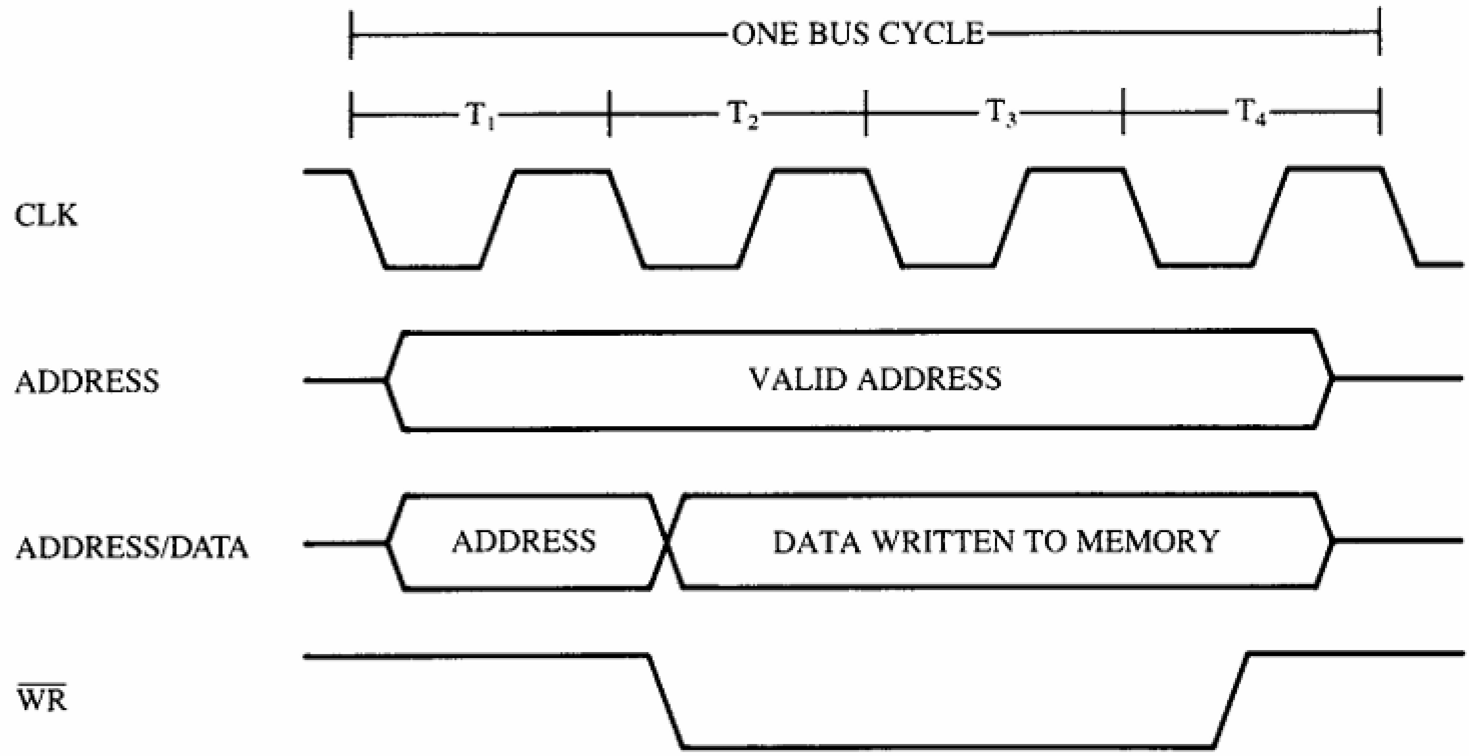
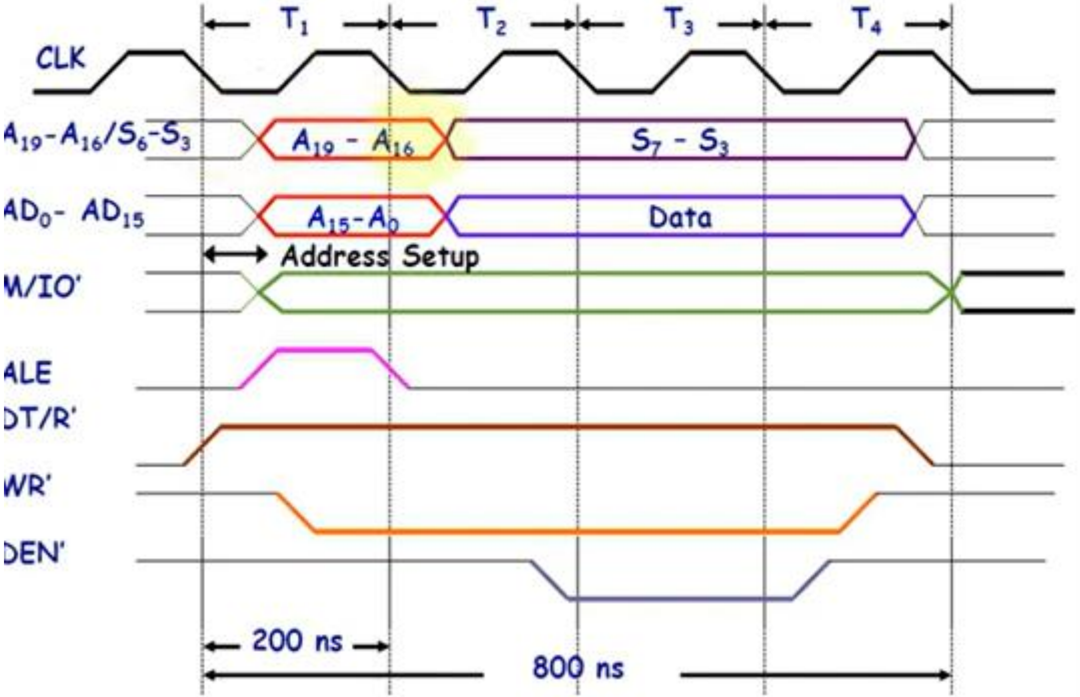
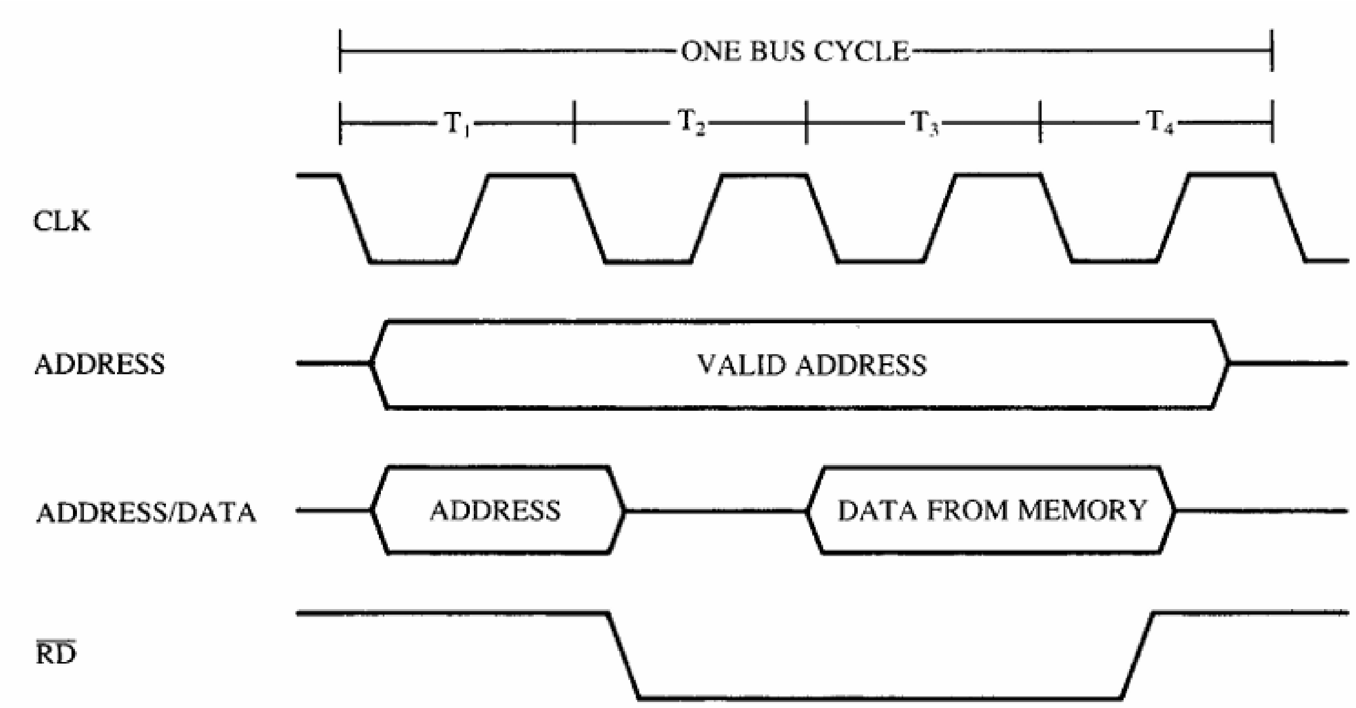
Timing Diagram Write operation



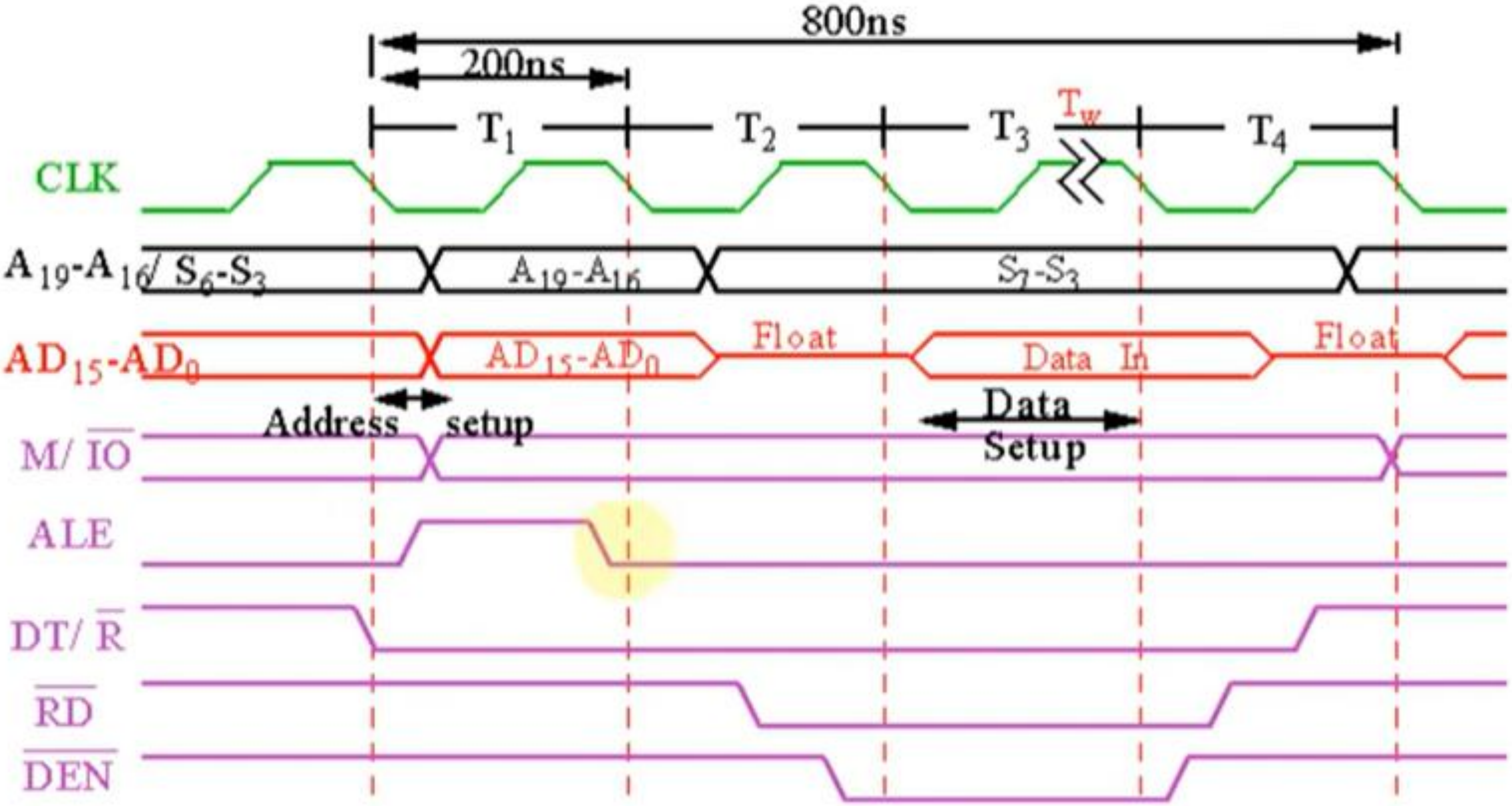
Memory Write Operation



Timing Diagram Read operation



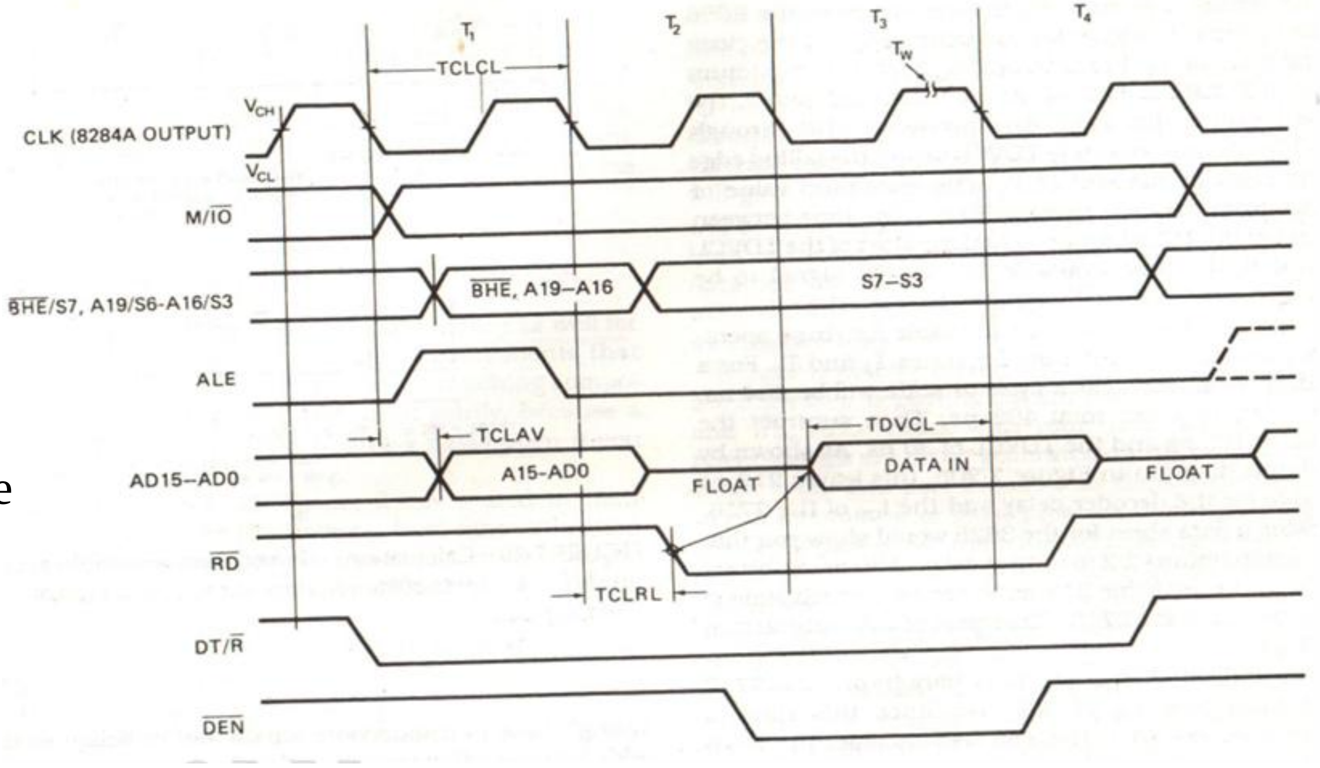
Memory Read Operation



Memory Access Time

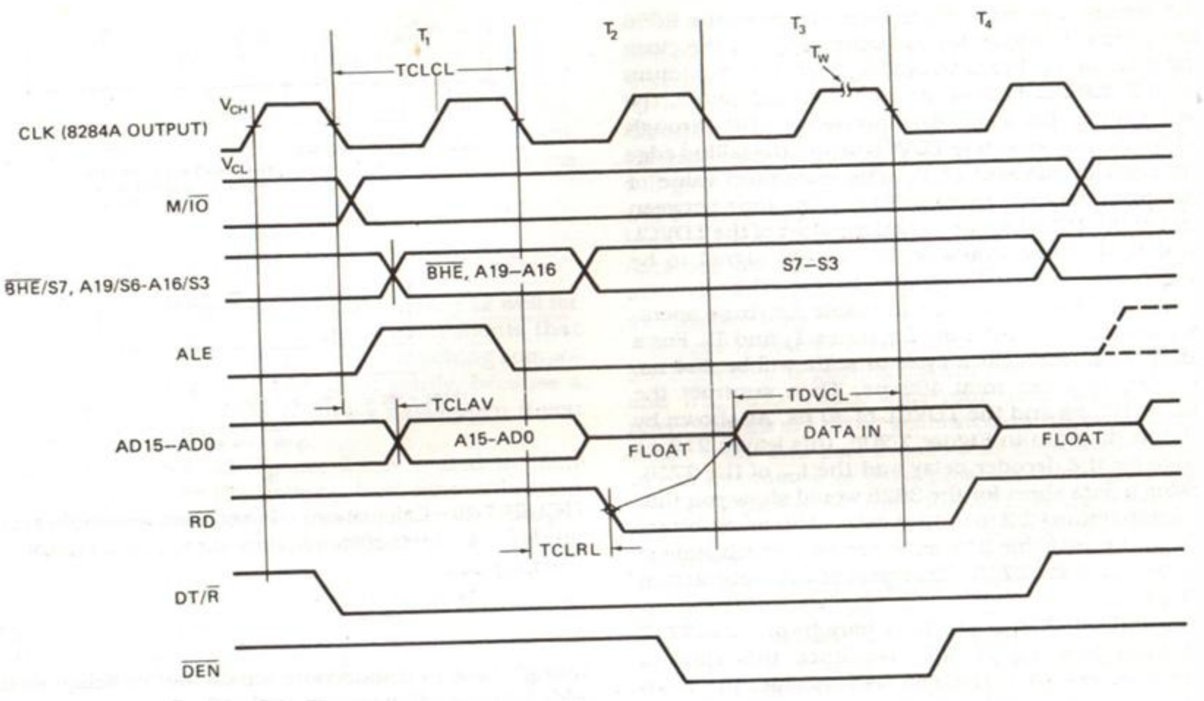
TCLAV- Time from Clock to Address Valid ; 110 ns for 5 MHz Clock

Memory Access Time : TCLAV time must be subtracted from the three clocking states (600 ns) separating the appearance of the address (T1) and the sampling of the data (T3).



Memory Access Time

TCLRL-Time from Clock to Read Line
TDVCL –Time Data Valid to Clock 30ns
Memory Access Time : 600 ns-110ns-30ns = 460 ns
A **wait state** (Tw) is an extra clocking period between T2 and T3 to lengthen bus cycle
On one wait state, memory access time of 460 ns, is lengthened by one clocking period (200 ns) to 660 ns, based on a 5 MHz clock.



Memory Types

- Two basic types:
- ROM: Read-only memory
- RAM: Read-Write memory

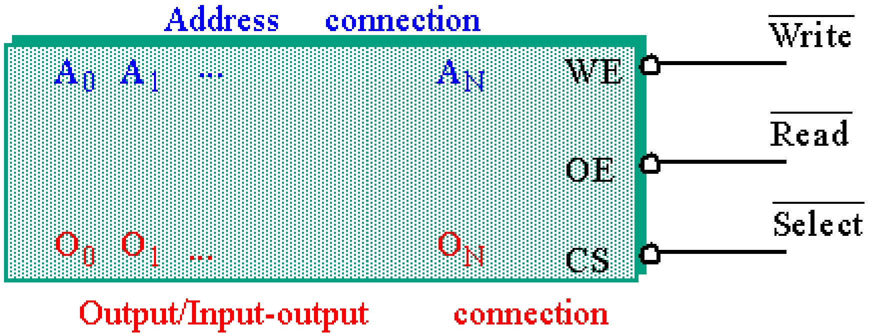
Four commonly used memories:

ROM

Flash (EEPROM)

Static RAM (SRAM)

Dynamic RAM (DRAM)



Memory Chips

The number of **address pins** is related to the number of **memory locations**.

Common sizes today are **1K to 256M** locations.

Therefore, between **10 and 28 address pins** are present.

The data pins are typically bi-directional in read-write memories.

The number of data pins is related to the size of the memory location.

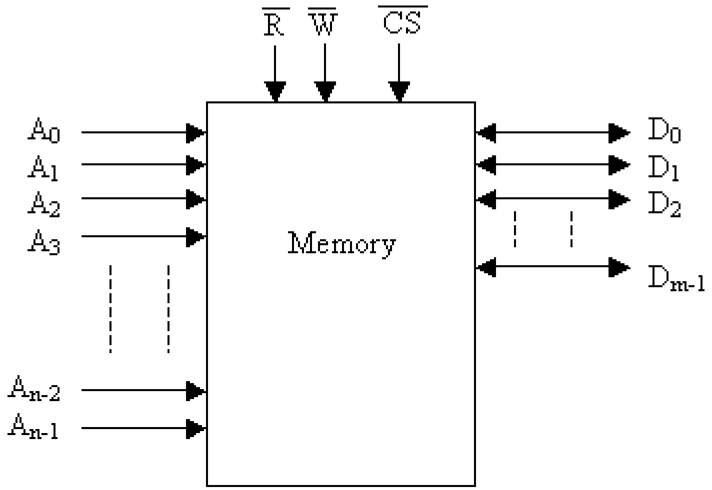
For example, an **8-bit wide (byte-wide)** memory device has **8 data pins.**

Catalog listing of **1K X 8** indicate a byte addressable **8K** memory.

Memory Chips

- Each memory device has at least one chip select **(CS)** or chip enable **(CE)** or select **(S)** pin that enables the memory device.

- This enables **read and/or write** operations. If more than one are present, then all must be **0** in order to perform a read or write.



Memory chips

Each memory device has at least one control pin.

For ROMs, an output enable (OE) or gate (G) is present.

The OE pin enables and disables a set of tristate buffers.

For RAMs, a read-write (R/W) or write enable (WE) and read enable (OE) are present.

For dual control pin devices, it must be hold true that both are not 0 at the same time.

ROM

Non-volatile memory: Maintains its state when powered down.

There are several forms:

ROM: Factory programmed, cannot be changed. Older style.

PROM: Programmable Read-Only Memory. Field programmable but only once. Older style.

EPROM: Erasable Programmable Read-Only Memory. Reprogramming requires up to 20 minutes of high-intensity UV light exposure.

Flash EEPROM: Electrically Erasable Programmable ROM. Also called EAROM (Electrically Alterable ROM) and NOVRAM (Non-Volatile RAM). Writing is much slower than a normal RAM.